MIL-D-60842(MU)
8 December 1970
SUPERSEDING
FA-PD-MI-2375, Rev. 1
25 July 1962

MILITARY SPECIFICATION
DELAY ELEMENT, CARTRIDGE: XM17
ASSEMBLY

1. SCOPE

1.1 This specification covers Delay Element, Cartridge: XM17, for use in cartridges for propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-13739	- Composition, Delay
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

D8596128	- Delay Element, Cartridge: XM17
C8826728	- Ammunition, Container, Metal, M21 for Delay Elements, M5 and M8
C8826729	- Box, Packing, Ammunition for Delay Elements, M5 or M8 in Metal Container, M21
IEL-8596128	- Inspection Equipment List for Delay Element, Cartridge: XM17 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016).

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The delay element assembly shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer or delay charge or derangement of any components.

3.4 Functional.

3.4.1 The delay element shall not fail to fire.

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3.4.2 Delay charge.-The delay charge shall comprise increments of Type I and Type II formulations shown in MIL-C-13739, in the proportions and quantities fixed by the contractor to produce the delay times required in Table I (see 6.3).

TABLE I

Temperature	Delay Time, Seconds (See 6.2)
-65° to -70°F	2.70 max.
65° to 75°F	2.25 ± 0.20
160° to 165°F	1.80 min.

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.-The part shall not have a burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection .-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

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4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 23 anodized units for protective finish tests and 30 cartridges for ballistics to be tested in accordance with Table II and 10 sets of unassembled components for visual and dimensional examination.

TABLE II

Paragraph Requirement	Paragraph Test	Test	Sample Size	Type of Test Destr.	Type of Test Non-Destr.
3.2	4.4.1	Protective finish	23	X	-
3.3	4.4.2	Vibration	12 (from Ballistics)	-	X
3.4.2	4.4.3	Ballistics	30	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Not Vibrated	Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
160° to 165°F	4	6

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-A lot shall consist of an identifiable group of metal parts assemblies and loaded with accepted delay charge from not more than one lot, propellant from not more than one lot and primers from not more than one lot.

4.3.2 Examination .-One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class the criteria in Table V shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used to perform visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed in ASA-B46.1 shall be used as a basis of comparison for surface roughness determination.

TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE V

Sample Size for Major Defects Number of Defects in Class	Total allowable defects per sample size			
	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14

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4.3.2 Table V (Cont'd)

Sample Size for Minor Defects Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 43	6	8	11	14
over 43	8	11	17	20

4.3.2.1 Body, Delay, Drawing C8594081, covering a detail of Drawing D8596128.

	Defect	Method of Inspection
CRITICAL		
1.	Diameter of flash hole	Gage
2.	Metal defective	Visual
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of body with pitch diameter of thread	Gage
104.	Diameter of thread relief	Gage
105.	Diameter of step at thread relief	Gage
106.	Diameter of cavity, max.	Gage
107.	Diameter of body, min.	Gage
108.	Depth of cavity, max.	Gage
109.	Length from cavity face to body diameter	Gage
110.	Thread damaged or not full	Visual
111.	Protective finish inadequate or defective	Visual
112.	Burr	Visual
113.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, max.	Gage
202.	Diameter of cavity, min.	Gage
203.	Diameter of body, max.	Gage
204.	Diameter of wrench holes	Gage
205.	Diameter of thread pilot	Gage
206.	Depth of cavity, min.	Gage

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
207. Depth of wrench holes	Gage
208. Total length	Gage
209. Length of thread pilot	Gage
210. Length of thread relief	Gage
211. Length of step at thread relief	Gage
212. Location of wrench holes	Gage
213. Radius or chamfer missing or incorrect	Visual
214. Surface finish improper	Visual

4.3.2.2 Gasket, Aluminum, Drawing B8594082, covering a detail of Drawing D8596128.

CRITICAL

None defined

MAJOR

101. Diameter, outside, min.	Gage
102. Diameter, inside, max.	Gage
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual

MINOR

201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Thickness	Gage
204. Surface finish improper	Visual

4.3.2.3 Primer Retainer, Drawing C8594083, covering a detail of Drawing D8596128.

CRITICAL

1. Diameter of flash hole	Gage
2. Depth of primer pocket	Gage
3. Metal defective	Visual

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4.3.2.3 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Pitch diameter of internal thread, max.	Gage
102.	Minor diameter of internal thread, max.	Gage
103.	Concentricity of head with body	Gage
104.	Concentricity of pitch diameter of thread with body	Gage
105.	Concentricity of primer pocket with body	Gage
106.	Concentricity of recess around primer pocket with primer pocket	Gage
107.	Symmetry of wrench flats with thread	Gage
108.	Diameter of head	Gage
109.	Diameter of primer pocket	Gage
110.	Diameter of body, min.	Gage
111.	Diameter of recess around primer pocket	Gage
112.	Diameter at base of angled recess	Gage
113.	Depth to bottom of thread cavity	Gage
114.	Depth of chamfer at primer pocket	Gage
115.	Depth of recess around primer pocket	Gage
116.	Length of effective thread, min.	Gage
117.	Location of datum diameter	Gage
118.	Thread damaged or not full	Visual
119.	Protective finish inadequate or defective	Visual
120.	Surface finish of primer pocket improper	Visual
121.	Burr or sharp corner	Visual
122.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, min.	Gage
202.	Diameter of body, max.	Gage
203.	Length of wrench flats	Gage
204.	Total length	Gage
205.	Width of wrench flats	Gage
206.	Location of wrench flats	Gage
207.	Radius or chamfer missing or incorrect	Visual
208.	Surface finish improper, except as otherwise classified	Visual

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4.3.2.4 Primer, Percussion, M42, Drawing B8799925 covering a detail of Drawing D8596128.

CRITICAL

1.	Improper assembly	Visual
2.	Any part missing	Visual
3.	Primer pellet broken, cracked or dislodged	Visual
4.	Metal defective	Visual
5.	Primer foil discolored	Visual
6.	Mixed primers	Visual

MAJOR

101.	Burr	Visual
102.	Foreign matter	Visual

MINOR

None defined

4.3.2.5 Assembly, Drawing D8596128.

CRITICAL

1.	Primer not seated to full depth of primer pocket	Visual
2.	Any part missing	X-ray/Visual 1/
3.	Improper assembly	X-ray/Visual 1/
4.	Marking missing, incorrect or illegible	Visual
5.	Mixed ammunition	Visual
6.	Flash hole missing	Visual
7.	Primer imcrimped or seal missing	Visual
8.	Delay charge not visible through flash hole	Visual

MAJOR

101.	Concentricity of head with body	Gage
102.	Diameter of body, max.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Total length, max.	Gage
202. Lacquer missing or inadequate	Visual

1/ The delay element shall be X-rayed with the element in a position horizontal to the rays of the X-ray machine. The negatives properly identified shall be forwarded to the procuring activity.

4.3.2.6 Ammunition Container, Metal, M21 for Delay Elements, M5 or M8, Drawing C8826728.

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat seal incomplete, loose or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.7 Box, Packing, Ammunition for Delay Elements, M5, M8, M14, M15, M16, M17 or XM28, Drawing C8826729.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.7 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-Twenty-three units of each anodized component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Vibration.-Twelve of the 30 assembled X-rayed delay elements of the initial production sample shall be subjected to vibration. After vibration, the 12 vibrated delay elements shall be X-rayed as directed in 1/ of 4.3.2.7. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the delay element, 12 additional delay elements shall be selected and tested as indicated herein.

4.3.3.3 Functional.

4.3.3.3.1 Ballistics.-The 30 test delay elements shall be temperature conditioned as indicated in Table VI.

TABLE VI

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

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4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8596128.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish tests shall be in accordance with MIL-C-60539. All anodized components shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Functional.

4.4.3.1 Ballistics.-The sample shall be grouped and temperature conditioned in accordance with 4.4.3.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.3.1.1.

4.4.3.1.1 Temperature conditioning.-A standard inert XM17 delay element, containing a temperature sensitive device shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the device contained in the inert delay element to the required temperature. The vent holes in the delay element shall be sealed against the environmental atmosphere of the conditioning chamber during the conditioning period. Upon completion of the temperature conditioning, the vent seal shall be removed.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings C8826728 and C8864729, except that all markings shall refer to Delay Element, Cartridge, XM17.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

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6.2 Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the venthole to the delay element.

6.3 Advisory delay charge composition (formulation).-Consist of three increments of charge as follows: approximately 6 grains Type I, 11 grains Type II and 6 grains Type I, charged in that order.

6.4 Supersession data.-This specification includes the requirements of Purchase Description FA-PD-MI-2375, dated 25 July 1962.

Custodian:

Army - MU

Preparing activity

Army - MU(FA)

Project No. 1377-A217

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R25

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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